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Problem

- Dense scaling raises training cost, activated compute and memory bandwidth; long-context decoding is dominated by KV-cache traffic.
- The target is a strong open model that is economical in both pretraining and inference.

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Scale & data

236B TOTAL, 21B ACTIVE

- DeepSeek-V2 is a mixture-of-experts model with 236B total parameters but only 21B activated for each token.
- It is pretrained on 8.1T high-quality multilingual and code tokens and extended to a 128K context.

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MLA

COMPRESS THE KV CACHE

- Multi-head Latent Attention projects keys and values into a compact latent vector shared across heads before reconstruction.
- The design retains multi-head expressiveness while sharply reducing per-token cache storage and memory reads.

DEEPSEEK-V2

DeepSeek-V2

DeepSeek-AI | arXiv:2405.04434v5

Combine sparse experts with compressed latent attention so a 236B-parameter language model activates only 21B per token and serves 128K context efficiently.

MLA

DeepSeekMoE

efficient inference

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DeepSeekMoE

FINE-GRAINED SPARSE EXPERTS

- Experts are segmented into smaller units, allowing more flexible combinations for each token.
- Shared experts capture common knowledge while routed experts specialize; auxiliary balancing prevents collapse.

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Efficiency evidence

- Versus DeepSeek 67B, training cost falls 42.5%, KV cache falls 93.3%, and maximum generation throughput rises 5.76x.
- After SFT and reinforcement learning, base and chat variants reach top-tier open-model performance.

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Meaning & limits

- DeepSeek-V2 shows attention memory and sparse activation must be optimized together, not only parameter count.
- The full model remains large to store and route; distributed MoE complexity, data opacity and safety evaluation remain concerns.